

# IT3081 – Statistical Modelling

## Group Assignment (10%)

### Industry Innovation Consulting Challenge: From Data to Strategic Decisions

#### 1. Assignment Overview

Modern organizations generate vast amounts of data every day. However, data alone does not create value. Organizations require professionals who can transform data into meaningful insights, evaluate alternative analytical approaches, and develop evidence-based strategies that support effective decision-making.

In this assignment, your group will act as an independent **Statistical Innovation Consulting Company**. Your consultancy team has been hired by an organization to investigate an important business or societal problem using statistical modelling techniques. Unlike a traditional university assignment, **no dataset will be provided**. Your team must independently identify an industry problem, locate a suitable dataset, conduct statistical analyses, evaluate advanced statistical methodologies, and prepare a professional consultancy report for the client.

The objective of this assignment is **not** to achieve the highest prediction accuracy. Instead, you are expected to demonstrate professional consultancy skills by combining statistical analysis, academic research, critical thinking, and industry knowledge to provide practical and evidence-based recommendations.

This assignment is designed to simulate the work of professional statistical consultants who support organizational decision-making through data-driven insights.

#### 2. Learning Outcomes

This assignment assesses the following learning outcomes.

**LO1:** Evaluate and apply advanced statistical inference, regression modelling techniques and generalized linear models.

**LO2:** Critically evaluate the suitability of experimental design methodologies.

**LO3:** Analyze multivariate relationships using Principal Component Analysis and discuss its applicability.

**LO4:** Evaluate Bayesian statistical methods and probabilistic approaches for organizational decision-making.

**LO5:** Critically discuss the applicability of time series analysis and stochastic modelling for future decision-making.

### **3. Group Formation**

- Students shall work groups with maximum 4 members.
- Each group will investigate a different industry problem.
- No two groups are permitted to work on the same problem.
- Every member is expected to actively contribute throughout the consultancy project.

### **4. Consultancy Scenario**

Your consulting company has been approached by an organization facing a strategic business challenge.

The organization possesses historical data but requires external experts to answer questions such as:

- What insights can be obtained from the available data?
- What statistical evidence supports important business decisions?
- Which predictive modelling technique is most appropriate?
- Are there alternative statistical approaches that should be considered?
- What strategic recommendations should management implement?

Your consultancy company has been hired to investigate these questions and prepare a professional consultancy report for the organization's senior management.

### **5. Selecting the Industry Problem**

Each group must identify one real-world business or societal problem from any industry.

Possible industries include, but are not limited to:

- Healthcare
- Banking and Finance
- Retail
- Marketing
- Education

- Agriculture
- Manufacturing
- Transportation
- Tourism
- Insurance
- Telecommunications
- Energy
- Government
- Environmental Management

Example consultancy problems include:

- Customer Churn Analysis
- Customer Purchase Behaviour Analysis
- Credit Risk Assessment
- Loan Default Prediction
- Hospital Readmission Analysis
- Disease Risk Prediction
- Product Quality Prediction
- Employee Attrition Analysis
- Student Performance Prediction
- Student Retention Analysis
- Crop Yield Prediction
- Road Accident Severity Prediction
- Sales Forecasting

The selected problem must be approved by the lecturer.

## **6. Dataset Selection**

Unlike previous assignments, your consultancy company is responsible for identifying and selecting an appropriate dataset.

Suitable sources include:

- Kaggle
- UCI Machine Learning Repository
- World Bank Open Data
- WHO Open Data
- Government Open Data Portals
- Data.gov
- Other publicly available repositories

The selected dataset should:

- Directly address the selected business problem.
- Contain sufficient observations for meaningful statistical analysis.
- Contain variables appropriate for descriptive, inferential and predictive analyses.
- Be publicly accessible.
- Be properly referenced.

Your report must clearly justify why the selected dataset is appropriate for solving the identified organizational problem.

## **7. Consultancy Tasks**

### **Task 1 – Understanding the Industry Problem**

Introduce the selected industry and business problem.

Your discussion should include:

- Industry background
- Organizational context
- Description of the business problem
- Importance of solving the problem
- Stakeholders affected

- Business decisions influenced by the problem
- Consultancy objectives

### **Task 2 – Research Landscape**

Conduct a comprehensive literature review to understand how researchers and organizations currently address the selected problem.

Your review should include:

- Current industry practices
- Existing statistical approaches
- Recent research findings
- Strengths and weaknesses of previous studies
- Research gaps
- Justification for your analytical approach

### **Minimum Requirements**

- Minimum 15 peer-reviewed research papers
- Minimum 10 journal papers
- At least 5 papers published within the last five years

The literature review should critically compare previous studies rather than simply summarizing them.

### **Task 3 – Dataset Understanding and Descriptive Analysis**

Perform a comprehensive descriptive analysis of the selected dataset.

This section should include:

- Dataset description
- Variable descriptions
- Data quality assessment
- Missing value analysis
- Outlier detection
- Descriptive statistics

- Appropriate visualizations
- Initial business insights

Explain every important observation from an organizational perspective.

#### **Task 4 – Statistical Inference**

Perform appropriate statistical inference techniques according to the nature of the selected dataset.

Where appropriate, apply:

- Comparison of Means
- Comparison of Proportions
- Comparison of Variances
- ANOVA

For every statistical analysis:

- Clearly state the hypothesis.
- Justify why the selected test is appropriate.
- Present the statistical results.
- Interpret the findings.
- Explain the practical implications for organizational decision-making.

Simply reporting statistical outputs is insufficient.

#### **Task 5 – Predictive Statistical Modelling**

Develop one or more predictive statistical models appropriate for the selected problem.

Possible techniques include:

- Multiple Linear Regression
- Logistic Regression
- Generalized Linear Models
- Ridge Regression
- LASSO Regression

- Elastic Net Regression
- Best Subset Selection
- Forward Selection
- Backward Selection
- Multi-response Regression

Your consultancy report should:

- Justify model selection.
- Compare alternative modelling approaches where appropriate.
- Evaluate model performance.
- Discuss assumptions.
- Identify strengths and limitations.
- Recommend the most suitable model for organizational use.

Prediction accuracy alone will not determine the quality of this section.

### **Task 6 – Critical Evaluation of Experimental Design**

Your consultancy company is **not required to conduct an experiment**.

Instead, critically discuss whether experimental design methodologies could be used to improve future investigations related to the selected business problem.

Evaluate the suitability of:

- Completely Randomized Design (CRD)
- Randomized Complete Block Design (RCBD)

Discuss:

- Situations where these designs may be useful.
- Expected advantages.
- Limitations.
- Implementation challenges.
- Practical feasibility.

Support your discussion using research literature.

### **Task 7 – Critical Evaluation of Principal Component Analysis**

Critically evaluate whether Principal Component Analysis (PCA) would be beneficial for the selected dataset.

Your discussion should include:

- Whether dimensionality reduction is necessary.
- Possible advantages.
- Limitations.
- Expected impact on predictive modelling.
- Situations where PCA should and should not be used.

Support your arguments using research findings.

### **Task 8 – Critical Evaluation of Bayesian Statistical Methods**

Evaluate the applicability of Bayesian statistical methods to the selected business problem.

Discuss:

- Naïve Bayes
- Bayesian Regression
- Bayesian Decision Making

Explain:

- Whether Bayesian approaches are appropriate.
- Potential advantages.
- Limitations.
- Situations where Bayesian methods may outperform traditional statistical approaches.

Support your discussion using published research.

### **Task 9 – Time Series Analysis**

You are **not required to perform time series analysis**.

Instead, critically discuss how time series analysis could be incorporated if historical time-dependent data were available.

Your discussion should include:

- Trend analysis
- Seasonal analysis
- Forecasting
- ARIMA modelling
- Possible business applications
- Expected organizational benefits

Support your discussion using recent literature.

### **Task 10 – Industry Innovation Proposal**

Consultants should not simply analyse problems. They should propose innovative solutions.

Based on your findings, design one innovative data-driven solution for the organization.

Examples include:

- Decision Support Dashboard
- Customer Intelligence Framework
- Risk Prediction Framework
- Resource Allocation Framework
- Early Warning System
- Data-Driven Decision Framework
- AI-assisted Analytical Workflow

Your proposal should explain:

- Business need
- Proposed solution
- Expected benefits
- Implementation challenges

- Required resources
- Organizational impact

Include a conceptual framework illustrating the proposed solution.

### **Task 11 – Industry Expert Validation**

Each consultancy team must consult at least one industry expert relevant to the selected domain.

Present your findings and obtain professional feedback regarding:

- Relevance of the problem
- Practicality of recommendations
- Implementation feasibility
- Organizational impact

Acceptable evidence includes:

- Interview summaries
- Meeting minutes
- Email communications
- Screenshots of online meetings
- Completed feedback forms

Discuss how the feedback influenced your final recommendations.

### **Task 12 – Final Consultancy Recommendations**

Prepare professional recommendations for senior management.

Recommendations should include:

- Strategic recommendations
- Operational recommendations
- Risk management considerations
- Ethical considerations
- Future research opportunities

Every recommendation must be supported using:

- Statistical evidence
- Research literature
- Industry expert feedback

## **8. Presentation and Viva (Group)**

Each consultancy team shall deliver

- 15-minute professional consultancy presentation
- 10-minute viva examination

Students should present themselves as consultants presenting recommendations to an organization's Board of Directors.

Every group member must actively participate in both the presentation and viva.

## **11. Assessment Structure**

| <b>Assessment Component</b> | <b>Weight</b> |
|-----------------------------|---------------|
| Consultancy Presentation    | 40%           |
| Viva Examination            | 60%           |
| <b>Total</b>                | <b>100%</b>   |

## **12. Bonus Opportunities**

### **Bonus Opportunity 1 – Real Industry Consultancy (+10%)**

Groups that complete this consultancy project for a **real company or organization** may receive up to **10 bonus marks**.

Requirements include

- Lecturer approval
- Real organizational problem

- Organizational engagement
- Presentation of recommendations to the organization
- Documentary evidence
- Organizational feedback

Bonus marks will be awarded based on the quality of industry engagement.

### **Bonus Opportunity 2 – Professional LinkedIn Showcase (+5%)**

Groups may receive up to **5 bonus marks** by producing a professional project showcase.

#### **Requirements**

- Maximum 5-minute video
- Explain the consultancy project, methodology, findings and recommendations
- Publish on LinkedIn
- Every group member must share the post
- Submit LinkedIn URLs before the deadline

The video should communicate the project to a professional audience.

### **13. Academic Integrity**

All datasets, research papers, software tools and Generative AI tools must be properly acknowledged.

The consultancy report must represent the original work of the group.

Any form of plagiarism, fabrication, falsification or academic misconduct will be handled according to the University's Academic Integrity Policy.

Responsible and transparent use of Generative AI is encouraged. Students remain fully responsible for the accuracy, originality and quality of all submitted work.